

OL7014 Integer Programming

Conference Report — Understanding a Talk Through the Optimization Lens

Due Nov 17, start of class

Pick **one** talk and describe its problem as an optimization model, in plain sentences (Korean or English) — no math required. **One page.** Submit one PDF to loss0f0706@gmail.com.

1. The talk

2–3 sentences

Title, speaker, session. Who are the decision makers? What problem are they trying to solve, and for whom? Why does it matter?

2. The problem as an optimization model

the core of the report

We decide *the decision variables — what gets committed? Which decisions are yes/no (binary), which are counts (integer), which are amounts (continuous)?*

to minimize / maximize *..the objective — the one number being improved. If the talk mixes several goals, pick the one you would optimize and say what happens to the others.*

subject to *....the constraints — what limits the decisions? budget or capacity, logic (if this, then that), rules and policy, physics of the system. Two to four constraints, each one sentence.*

from the data *what must be known before the model can be written down — costs, demands, distances, a network?*

3. Analysis

4–6 sentences

Is their problem genuinely challenging from an optimization perspective — and what makes it so (discrete decisions, scale, uncertainty, competing objectives)? Do you see a mismatch between their actual problem and the optimization model they propose, and how would you improve their approach? Do they consider deterministic data that should have been random — would you consider stochastic data instead? Is there a place where sharper modeling — integer decisions, an explicit objective — would beat their solution?